

IMPACT CRATERS

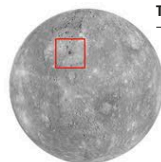
Mercury is covered with impact craters ranging in size from small, bowl-shaped craters to a basin that is a quarter of the diameter of the planet. The relative ages of different surface areas can be deduced because older regions have a greater density of superposed craters and because older craters become degraded by subsequent impacts distributing fragmental material (regolith).

SOUTH POLE

The temperature is permanently freezing in Mercury's south polar region because it receives very little sunlight.

SHAKESPEARE REGION

Caloris Basin



TYPE Impact crater

AGE 4 billion years

DIAMETER 960 miles (1,550 km)

MESSENGER MAP

Caloris is the biggest well-preserved basin on Mercury, larger than the state of Texas. Shortly after being formed by a major impact about



3.9 billion years ago, its floor became flooded by lavas that show up as orange in exaggerated-color images, in contrast to the generally darker and bluer terrain outside the basin. Subsequent smaller impacts into the basin penetrated through the lavas, exposing very dark blue material from the original floor of the basin.

There are no "ghost craters" (see Agwo Facula, opposite) on the Caloris floor. This indicates either that the depth of lava flooding was so great that ghost craters are entirely obscured, or that there was not enough time for craters to be made on the floor of the new basin before the lavas were erupted.

Bright orange spots just inside the southern rim of the basin are thought to be material erupted by explosive volcanic eruptions, such as Agwo Facula (see opposite).

The name "Caloris" comes from the Latin for heat, bestowed because the Sun is overhead at this longitude during alternate perihelia, so midday temperatures here are particularly high.

BASIN FLOOR

The lavas on the basin floor are wrinkled by ridges and cut by fractures, which reflects a history of lava cooling and subsidence and scarring by impact craters of various sizes.

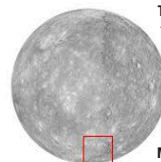
IMPACT BASIN

This exaggerated-color image shows that the lavas covering the basin floor (orange) have a different composition to the surrounding area.



BEETHOVEN REGION

Bach Crater



TYPE Impact crater

AGE 4 billion years

DIAMETER 133 miles (214 km)

MESSENGER MAP

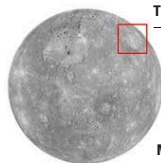
This two-ringed basin represents an intermediate class of craters, between the slightly smaller ones with central mountainous peaks and larger ones with multiple rings. The prominent inner ring is half the width of the outer, and the overall circularity is impressive. Bach was formed toward the end of the period of heavy bombardment. Lava later flooded the crater, producing the smooth floor.



TWO-RINGED BASIN

SHAKESPEARE REGION

Degas Crater



TYPE Impact crater

AGE 500 million years

DIAMETER 37 miles (60 km)

MESSENGER MAP

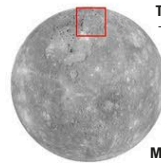
The flat floor of this relatively young crater appears to be covered by a layer of impact melt that cracked as it cooled. The central peak appears bright because volatile substances have migrated there to form hollows (see opposite). Much of the ejecta surrounding the crater's rim is dark, so-called "low reflectance material" excavated by the crater-forming impact and probably originating from a graphite-rich subsurface layer.



LOW-REFLECTANCE MATERIAL

SHAKESPEARE REGION

Brahms Crater



TYPE Impact crater

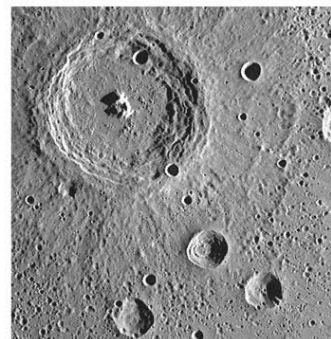
AGE 3.5 billion years

DIAMETER 60 miles (97 km)

MESSENGER MAP

A large, mature, complex crater to the north of the Caloris Basin, Brahms has a prominent central mountainous peak about 12 miles (20 km) across. The walls have slipped inward, forming a series of elaborate, concentric, stairlike terraces and a highly irregular rim. This structure is typical of a crater this size—craters with diameters less than 6 miles (10 km) are bowl-shaped, and craters with diameters

greater than 80 miles (130 km) develop central rings (see Bach, above). Radial hills of ejecta surround Brahms.



CENTRAL PEAK

This 2-mile (3-km) high mountain was produced when the subsurface rock rebounded after being struck by an asteroid.